

BIOLOGICAL METHODS OF ANALYSIS

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Introduction

- Microbial quality control is conducted on pharmaceuticals and medical devices.
- Reasons for quality control(QC)
 - Health significance
 - Economic significance.
- There are different biological methods which are used for microbial QC
 - ✓ Pyrogen Test
 - ✓ Microbial Limit Test
 - ✓ Sterility Test
 - ✓ Preservative Efficacy test
 - ✓ Microbiological Assay

Introduction....

1. Pyrogen Test

- ❖ Pyrogens / Endotoxins/ are defined as substances that when injected in sufficient amount gives rise to a variety of responses and symptoms.

- **Physiological Effects of pyrogens**

- | | |
|-----------------------------|--------------------|
| ✓ Increase body T° | Papillary dilation |
| ✓ Decrease in respiration | Head ache |
| ✓ Sever diarrhea | Renal/liver damage |

- ❖ All microbes appear to be capable of producing pyrogens but the most potent are associated with *gram negative bacteria*
- ❖ Endotoxins...consists of a lipopolysaccharide (LPS)
- ❖ Lipopolysaccharides are **thermostable** and fairly *less sensitive to weak acid and weak base*
- ❖ Pyrogens are produced in the cell wall of both living and dead microbial

Methods of pyrogen test

1. Rabbit test (In-vivo test)

- ❖ The product is administered to the rabbit and the change (increase) in temperature is measured in a specific period of time

Why rabbit?

- In expensive
- Easy to handle
- Rabbits show same response to pyrogens as human on per kg body weight basis

Limitation: - rabbits frequently produce false positive
Response

Rabbit test ...

Test Procedure

- Three healthy rabbits are selected
- Accurate temperature (rectal) is initially taken by a thermometer.
- Test solution is warmed to $37^{\circ}\text{C} \pm 2^{\circ}\text{C}$ and administered,
- Rectal temperatures are detected every 30 min for 1-3 hrs.
- The product passes *if no rabbit shows a temperature raise of 0.5°C or more,*
- If any rabbit shows $> 0.5^{\circ}\text{C}$ raise, the test is repeated on 5 more rabbits
- Now, if not more than 3 of the eight rabbits show a raise $> 0.5^{\circ}\text{C}$ and if the sum of the eight individual rabbits increase is not greater than 3.3, *then the product passes.*

- Advantages – detect all types of endotoxins
- Disadvantages
 - Time taking and development of tolerance by rabbits.
 - Interference from some pharmaceuticals.
 - Response of rabbit depends on time
/afternoon/

2. In-vitro pyrogen testing

- Limulus Amoebocyte lysate (LAL) Test
- A test for the determination of gm-negative bacterial endotoxin
- It employs a lysate Protein obtained by lysing the blood cells (amoebocytes) of the horseshoe crab
- The lysate protein is the most sensitive substance known for endotoxins.

A. Gel-clot method - performed using equal volume of the product and the reagent.

- Reaction is carried out in a test tube
- Reaction mixture left for 1hr
- After one hour there might be a gel formed
- Invert by 180°
- After inverting if there remains an in fact gel +ve
- If all the contents collapse and flow upon inverting -Negative (-ve) result

B. Chromogenic method

- LAL reagent is originally prepared by mixing with peptides which can form coagulin
- LAL does not act on the peptides in the absence of endotoxins

Result - Coagulation -+ve
 - Otherwise - -ve

C. Turbid metric method

- The same as the chromogenic method
- Turbidity is measured
- Is quantitative method

Methods of Depyrogenation

Different methods can be employed

- Heat 250°C – 300°C for 30 min-2 hr provided that our product is thermostable
- Acid-base hydrolysis – PH – (acidity /basicity/ should be very strong)
- Oxidation of the pyrogens using H_2O_2 , HNO_3 , O_3
- Ionization ionizing with radiation /
- Ultrafiltration and Distillation

2. Microbial limit test

- ✓ A method used to determine permissible level microbes in non- sterile products
- ✓ In general, there are 3- steps employed in the limits
 - Predatory test
 - Actual /TAC/ test
 - Confirmatory test

Microbial limit preparatory test

- ✓ Determine whether the product have antimicrobial activity or not.
- ✓ Because if it have, it may suppress the growth of microbes.
- ✓ If it contains antimicrobial activity the product should be removed or inactivated before the limit test is carried out.
- ✓ Filtration or Dilution
- ✓ Neutralization

Microbial limit test

A. Total aerobic count (TAC)

- is a test for bacteria and fungus

Methods – 4 types.

- ✓ Membrane filtration method
 - The suspension containing the microbes is filtered there by making the microbes to be retained on the membrane (filter).
 - The membrane will be put in a growth media and incubated.
 - Number of colonies are counted
 - After 5 day, no of colonies will be count
- Bacteria - no. of colonies allowed are 300 colonies/plate
- Fungus - no. of colonies allowed are 100 colonies/plate

✓ Pour plate method

- In this method the product is mixed with melted media
- Solidify at $T^{\circ} < 45^{\circ}\text{C}$ (usually 37°C)
- Incubate for 5 days
- Count number of colonies
- Results – interpret as above (similar to the 1st method)

✓ Spread plate method

- Solid media is prepared out which the product will be spread on the surface of the media.
- Incubate for 5 days
- Result interpretation as above

- ✓ Serial dilution method(Probable number method)
 - 12 test tubes with the same shape and size are required
 - The test tubes are grouped in to 4
 - Add to the first group 10ml of sterile media
 - Add 9ml of the sterile media to group 2,3,4
 - Add 1 ml of the product to group 2.
 - Take 1 ml of the mixture from group 2 and add to the 3rd group (1st of G_2 to 1st of G_3)
 - Same is done to group 4 (from G_3 to G_4)
 - Incubate for 5 days
 - Count colonies in each tube
 - If this is not possible 0.1ml of each group is taken, added to plate-incubate & count colonies.

Results- Evaluate the product (Interpret the result) according to the information given in pharmacopoeia.

B. Test for Designated microbes

✓ According to USP, the four major species of concerns are Escherichia coli (E-coli)

- Its presence is an index of fecal pollution of water.

Staphylococcus aureus

- Human nose, throat and skin
- Workers with poor sanitary quality could be the source of product contamination.

Pseudomonas aeruginosa

- Universally distributed microbe.
- Agent of many plant and animal disease.
- Highly resistant to antibiotics.

Salmonella

- Common enteric micro flora of human and other animals
- Responsible for enteric fevers and
- Its presence is possibly an indication of
- Sewage contamination of the product
- Low hygienic quality

Fungus...Candida Albicans and Asparagus Niger

Specific tests

- Performed to rule out the presence of the above designated species which are required to be absent
- Specific /selective / growth / medias are used for the tests.

3. Sterility Test

❑ Sterility refers to an absolute condition that is taken to mean the absence of any viable no that can reproduce under fixed conditions.

➤ Two methods of sterility test

A. Immersion (direct inoculation)

- Prepare the appropriate media
- Inoculate with the product
- Incubate for 14 day
- Observe if there is a growth if yes-reject

Advantage — requires small volume of sample

- Can be used for non-liquid articles such as creams ointments or medical devices.

Limitation - Cannot be used for those products with antimicrobial activity.

B. Membrane filtration method

- Test product is made to pass through a filter so that organisms of interest are retained in the filtering material
- Incubate the filtered microbes for 14 days and check growth

Advantages - large volumes can be analyzed

- used for products with antimicrobial activity.

Limitation - used only for solutions.

- requires large volume of sample.

✓ Interpretation of results – media examined on the 3rd, 4th, 5th, 6th, 7th, 8th and 14th day.

• Controls should be there Positive control

Negative control

3. Antimicrobial preservative efficacy test

- ✓ Evaluates the effectiveness of antimicrobial preservative
- ✓ Product is challenged with standard strains of microbes
- ✓ After a period of incubation the ability of the preservative employed to control the multiplication of the added microbes is verified.

Test organisms employed

- ✓ Standardized with respect to BP, USP and EP
- ✓ Bacteria – *S. aureus*, *E-coli*, *P. aurognosa*
- ✓ Yeast - *C.albicans*
- ✓ Mold – *A. niger*.

❑ Reasons for selecting the above microbes

- ✓ They are representatives of the group which are typically contaminating during manufacturing
- ✓ They arise as contaminants during use.
- ✓ Example- the skin of patients can be a source for *S.aureus* and *C-albicans*.
- ✓ They have undemanding nutritional requirements and commonly associated with preservative resistance.
- ✓ They have pathogenic potential.

❑ For preservative efficacy test, 100-150ml organ of product is required The test is carried out for 28 days.

- ✓ High concentration of Inoculum (10^7 to 10^8 cells (ml))
- ✓ The microbes should be with in log phase (similar property of the organism)

Interpretation_

- ✓ Results are examined at 7th, 14th, 21st, 28th days and any change in the appearance is recorded.
- ✓ The concentration of viable bacterium is reduced to not more than 0.1% of the initially count by the 14th day.
- ✓ Fungal SPP. Remain at the same or below the initial number at the 14th day.
- ✓ The concentration of each test organism remains at or below those designated level (pharmacopeia's specification) during the reminder of the 28 days of test period.

4. Microbiological Assay

- ❑ Requires fulfillment of 3 factors:-
 - Test compound /Antibiotic/
 - Reference standard
 - Sensitive reference strain (MO)

Principle — it is based on the comparison of the effects of the test a suitable organisms in a nutrient medium.

- ❑ The test MO, should
 - Be sensitive to the antibiotic
 - Slow graded response: successive dilution should show decrease or increase in effect based on concentration.
- ❑ Ideally reference standard
 - Available in sufficient quantity
 - Highly stable.
 - Qualitatively be identical with the test compound that is being assayed.
 - Be preferably a simple substance.

Two methods

A. Plate or cylinder/Diffusion/ Assay

- Keep ready sterile glass plates
- Pour the medium into plates and solidify
- Drop cylinders of 8mm- 6 mm diameter from height of 12 mm and use Cylinders per plate
- Add 100 l of test & reference Solution to the plate.
- Keep in refrigerator for 1-2 hrs ---diffusion of antibiotic
- Incubate at 32-35°C for 12-18 hrs.
- Observe the zone diameter /Inhibition zone/ and measure the diameter.

B. Tube method

- Sterilized, equal size tubes are required
- To each test tubes 1 ml of test product and 9 ml of broth medium is added (for both reference & test product)to each test tubes, specified number of microbes are added.
- After 4 hrs of incubation, the growth of the microbes will be inhibited by heating 80°C .
- Measure turbidity
- The lower the turbidity, the potent the product is